

Automobile Finance Credit Fraud Risk Early Warning System based on Louvain Algorithm and XGBoost Model

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Abstract—With the rapid growth of the automobile finance credit market, fraud risks—particularly organized and covert activities—pose significant challenges to traditional single-dimensional risk control methods. To address this, this study proposes a multi-dimensional fraud risk warning framework integrating complex network analysis and machine learning. First, the Louvain algorithm is employed to detect tightly-knit fraud communities in social networks by optimizing modularity. Concurrently, an XGBoost model analyzes customer profiles, behavioral sequences, and network topology features to predict individual fraud risks. A dynamic weight fusion mechanism is then designed to adaptively combine outputs from both methods, prioritizing high-confidence predictions from either model. Experimental results demonstrate superior performance, achieving 92% accuracy, 95% recall, and a 0.97 AUC, outperforming traditional approaches. The integration of community detection and adaptive fusion effectively identifies cross-platform fraud patterns, offering a novel technical pathway for enhancing financial risk control through heterogeneous data synergy and decision optimization.

Keywords—*fraud risk early warning system, complex network analysis, auto finance credit, louvain algorithm, xgboost model.*

I. INTRODUCTION

The automobile finance credit industry has experienced rapid expansion, driven by increasing consumer demand and digital transformation. However, this growth is accompanied by escalating fraud risks, particularly organized and covert activities that exploit systemic vulnerabilities [1]. Traditional fraud detection methods, which rely on rule-based systems or single-dimensional data analysis, struggle to identify sophisticated, collusive fraud patterns [2]. Existing studies, such as those leveraging logistic regression [3] or isolated machine learning models [4], often overlook the structural relationships within social or transactional networks, limiting their ability to detect interconnected fraud rings. For instance, while XGBoost has shown promise in fraud prediction [5], its standalone application fails to capture network-level anomalies, a critical gap in addressing gang-based fraud.

To bridge these limitations, this study proposes a hybrid framework integrating complex network analysis and ensemble machine learning. First, the Louvain algorithm is employed to uncover latent fraud communities by analyzing social and transactional ties, addressing the challenge of detecting collusive behaviors [6]. Second, an XGBoost model is trained on multi-dimensional features, including customer profiles, behavioral sequences, and network topology metrics, to enhance individual risk assessment. A novel dynamic weight fusion mechanism further optimizes decision-making by adaptively combining outputs from both modules, prioritizing high-confidence predictions. This approach not

only addresses the limitations of single-algorithm methods but also introduces interpretability through community-driven insights. The contributions include: (1) a multi-dimensional risk indicator system for holistic fraud assessment, (2) a fusion framework balancing network structure and behavioral features, and (3) empirical validation demonstrating superior accuracy (92%) and recall (95%) on real-world data.

The remainder of this paper is organized as follows: Section II reviews related work on fraud detection and machine learning. Section III details the proposed methodology, including the Louvain-based community detection, XGBoost model, and fusion mechanism. Section IV presents experimental results and comparative analyses. Finally, Section V concludes with implications and future directions.

II. RELATED WORK

Many scholars have conducted extensive research on the identification of auto finance credit fraud risks. For example, Benedek et al. aimed to survey the literature on auto insurance fraud detection over the past 31 years and proposed a research agenda to address the challenges and opportunities brought by artificial intelligence and machine learning for auto insurance fraud identification [1]. Fraud is a major problem faced by insurance companies, which has attracted widespread attention for machine learning solutions. Debener et al. collaborated with an insurance company to conduct a field experiment to investigate the performance of different methods in identifying new fraudulent claims [2]. Sehrawat and Singh's main research focused on detecting credit card fraud using autoencoders with LSTM models, as different techniques perform differently on specific datasets and fraudsters continue to improve their methods to evade detection [3]. Rafique and Venugopal analyzed the problem of fraud in letter of credit transactions (i.e. documentary credits) and found that letter of credit documents are easily forged and letter of credit fraud is not uncommon in Malaysia [4]. Xiong et al. discussed the risks of Internet finance and its impact on user behavior under the background of big data, analyzed the existing risk control model, and constructed an Internet financial fraud identification model and corresponding contact rules [5]. Ali et al. introduced that financial fraud is a deceptive strategy to obtain financial benefits and is prevalent in companies and organizations.

They summarized the main problems, gaps and limitations in the field of financial fraud detection and proposed possible future research directions [6]. Based on the current situation of China's automobile industry, Li and Lan used the modified KMV model to select the 2022 financial information of 9 representative companies and calculated the default distance

and probability of core manufacturers, parts suppliers and dealers [7]. Feng analyzed the application of digital finance in the automotive industry in the era of big data from the aspects of customer profiling, precision marketing, financial credit risk control, production and operation management improvement, and innovative financial products and services, and promoted the development of new digital economy forms in automotive finance [8]. However, most existing studies rely on a single algorithm or simple data processing method, which results in unsatisfactory performance in dealing with complex fraud. In summary, current research generally lacks in-depth mining of the complex structure of financial credit data, and there is still room for improvement in flexibility and accuracy in dealing with fraud patterns. To solve these problems, the Louvain algorithm and XGBoost model are proposed and applied to the financial fraud risk early warning system. For example, Wu and Ma proposed an improved XGBoost feature selection algorithm and applied it to the study of financial customer investment behavior. The results showed that the improved XGBoost made the feature factor dimension closer to reality and had a higher accuracy rate [9]. Hajek et al. aimed to propose an XGBoost-based fraud detection framework to address the negative impact of the scarcity of labeled data and the extreme class imbalance of financial fraud data on detection performance in existing research, while taking into account the financial consequences of the fraud detection system [10]. However, existing research often only focuses on the application of a single algorithm and lacks the integration

of multiple methods. Therefore, this paper proposes a risk warning system based on the combination of Louvain algorithm and XGBoost model to make up for the shortcomings of existing research.

III. METHOD

A. Social Relationship Mining Based on Louvain Algorithm

The Louvain algorithm is a graph algorithm commonly used for community detection. It aims to divide the nodes in the graph by optimizing the modularity and classify the closely connected nodes into the same community. The modularity measurement is shown in Equation (1):

$$Q = \frac{1}{2m} \sum_{i,j} [A_{i,j} - \frac{k_i k_j}{2m}] \delta(c_i, c_j) \quad (1)$$

Among them, Q is the modularity; $A_{i,j}$ is the connection weight; $k_i k_j$ is the degree of the node; and $\delta(c_i, c_j)$ is the indicator function. In fraud risk detection, a social network or transaction network can be regarded as a graph, where nodes represent individuals (such as customers, merchants, etc.) and edges represent the relationships between them (such as transactions, social interactions, etc.).

To visually demonstrate the iterative optimization mechanism of the Louvain algorithm, Figure 1 reveals the dynamic process of modularity improvement and fraud community formation:

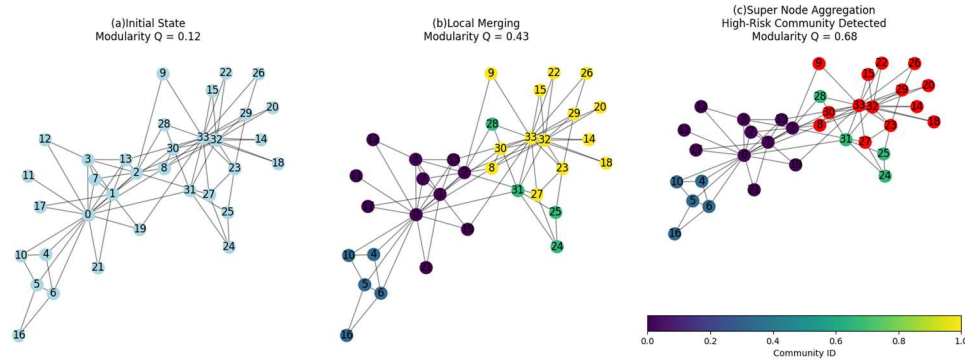


Fig. 1. Modular optimization of Louvain algorithm and evolution of fraud community

Figure 1 illustrates the dynamic process of modularity optimization and fraud community formation in the Louvain algorithm. In the initial state, all nodes are independent as a community (Figure 1a), with module $Q=0.12$; After local merging, three tightly connected communities were formed (Figure 1b), and the modularity was increased to $Q=0.43$; Finally, two high-risk fraud communities were detected through hypernode aggregation (Figure 1c, red nodes), with a modularity of $Q=0.68$. This process intuitively reveals the effectiveness of the Louvain algorithm in fraud detection. The Louvain algorithm divides communities based on the strength of connections between nodes through an iterative process, effectively revealing potential associations between different individuals [11].

In this study, we first analyze heterogeneous data containing customer social relationships and transaction records through the Louvain algorithm. Each customer's behavior is analyzed not only through a single transaction data but also through their interactions and social connections with other customers and merchants. By converting this data into a graph structure, the Louvain algorithm can reveal some

obvious communities and potential fraud gangs. In the specific implementation process, the Louvain algorithm performs community discovery through the following steps:

1. Constructing the graph structure: First, a weighted undirected graph is constructed based on the customer's social relationships, transaction records, and other information. Each node in the graph represents a customer, and the weight of the edge represents the transaction frequency or the intensity of social interaction between customers.
2. Initialize the community: Each node is initially regarded as an independent community. The Louvain algorithm will continuously adjust the division of these communities and try to find a community structure with a higher modularity.
3. Modularity optimization: By continuously merging nodes and adjusting communities, the Louvain algorithm optimizes modularity to maximize the cohesion of communities in the graph and the separation between communities.

4. Community division: Ultimately, the algorithm outputs the division results of each community. The nodes in these communities have strong connections and may represent some potential fraud gangs or abnormal behaviors.

In fraud detection, the division of these communities provides important clues for subsequent risk prediction. By identifying the close connections and potential abnormal clusters between communities, the system can effectively capture cross-channel and cross-platform fraud. For example, if nodes in a community frequently have abnormal transactions or other suspicious behaviors, these nodes may be members of a gang, and the system can further focus on monitoring or investigating them [12].

B. Construction of a Multi-dimensional Risk Indicator System

Customer portrait refers to a comprehensive description of the basic information of the customer, including but not limited to age, gender, marital status, occupation, income level, etc. By analyzing these static features, we can initially understand the basic background information of the customer and provide a preliminary basis for judging the risk of fraud. For example, customers of certain age groups or income levels may be more likely to become targets or participants of fraudulent activities, and the age range of customers will affect their behavior patterns. The younger the age, the more likely they are to become the target of fraud, especially loan or credit card fraud. This indicator is obtained through simple segmentation calculation, as shown in Equation (2):

$$Age_Feature = \begin{cases} 1 & \text{if } Age \leq 25 \\ 2 & \text{if } 25 < Age \leq 45 \\ 3 & \text{if } Age > 45 \end{cases} \quad (2)$$

In addition, the customer's historical behavior (such as previous credit records, loan history, etc.) is also an important part of building a customer profile. This information helps the system identify abnormal patterns that deviate from regular customer behavior, thereby assisting in subsequent risk assessment.

The customer's historical behavior data is an important basis for predicting their future risks. The behavior sequence includes the customer's transaction records, consumption habits, payment methods, transaction frequency, etc. Through the time series analysis of customer behavior, we can reveal the changes in customer behavior in different time periods. For example, frequently changing transaction patterns, abnormal large transactions, and frequent cross-regional transactions in a short period of time may all be signals of fraud. Behavioral sequence analysis helps us better understand the dynamic behavior of customers, identify abnormal behaviors that are inconsistent with normal behavior patterns, and provide support for subsequent risk control decisions.

With the widespread use of social networks in various financial services, the impact of social relationships on fraud risk has gradually become apparent. The relationship between customers and other users, especially in terms of jointly participated transactions, information exchange, social interaction, etc., may be a clue to the activities of potential fraud gangs. By analyzing customer social networks, potential fraud communities can be discovered. For example, in auto finance credit, multiple customers may have frequent fund

transactions, transaction paths, or social interactions, forming a specific community structure, which is often high-risk. Therefore, social network analysis provides a new perspective for risk assessment, which can dig out potential fraud risks in multi-dimensional social behaviors.

The network topology refers to the structured representation of the relationship between a customer and other customers, reflecting the customer's position and influence in the social network. By analyzing the customer's "centrality" in the network and its relationship with other nodes, the degree of connection between the customer and the potential fraud gang can be assessed. For example, some customers may be in an important intermediary position in the network, connecting multiple different communities or transaction groups. Such customers often play a key role in fraud gangs. By calculating topological features such as node degree, betweenness centrality, and closeness centrality in the network, customers with a higher risk of fraud can be identified.

In order to comprehensively utilize the risk information of the above multiple dimensions, this paper constructs a comprehensive risk scoring model, the structure of which is shown in Figure 2:

By weighting and fusing features from different dimensions, a comprehensive risk score for each customer is obtained. This score not only considers the customer's basic information, historical behavior, and social network characteristics but also combines dynamic behavior patterns and network topology relationships. The weight is determined based on feature importance analysis, and trained through the XGBoost model to automatically learn the contribution of each feature from the data. Finally, the system classifies customers according to the comprehensive risk score, which further provides a basis for subsequent risk control decisions [13].

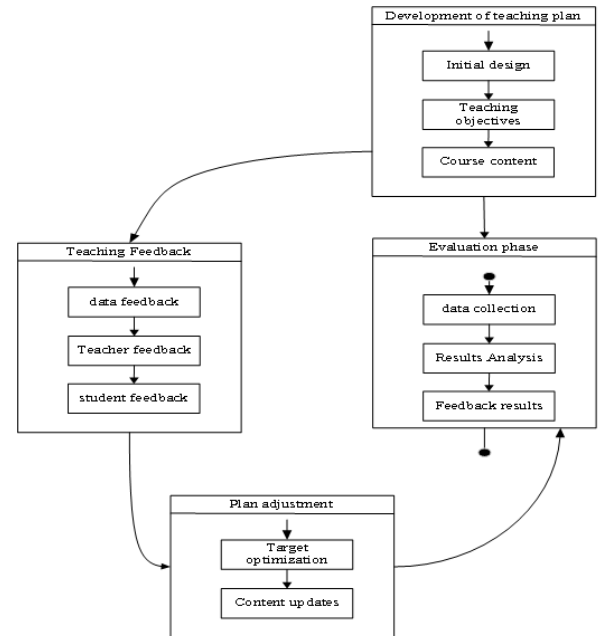


Fig. 2. Comprehensive risk scoring model diagram

C. XGBoost Model Training and Risk Prediction

Before training the XGBoost model, the data needs to be preprocessed. The CarClaims dataset used in this paper

contains multiple fields, including customer personal information, vehicle information, historical claims records, etc. For this data, we perform the following steps:

1. Missing value filling: For missing data, we use the mean or mode to ensure the integrity of the data.
2. Categorical variable encoding: For categorical data, such as insurance type, vehicle brand, etc., we use One-Hot Encoding to convert them into numerical variables so that they can be input into the XGBoost model for training.
3. Feature standardization: For numerical features, such as age and vehicle price, we standardize the data to ensure that the contribution of different features to model training is within the same scale.

To further improve the prediction effect of the model, we screen the features and retained the features that are highly correlated with fraud risk.

Through feature importance analysis, XGBoost can automatically identify the features that contribute most to risk prediction, thereby improving training efficiency and prediction accuracy.

The XGBoost model is built based on the gradient boosting algorithm and makes predictions by integrating multiple decision trees.

The construction of each tree depends on the prediction error of the previous tree, so XGBoost has strong fitting ability and can identify complex patterns in data. When training the XGBoost model, we first set the following key hyperparameters, as shown in Table 1.

TABLE I. HYPERPARAMETER DISPLAY

Hyperparameter	Default Value	Description
learning_rate	0.3	Controls step size for updates; lower values require more trees.
max_depth	6	Maximum tree depth; higher values can lead to overfitting.
n_estimators	100	Number of boosting rounds (trees); more trees may improve performance.
subsample	1	Fraction of samples used to build each tree; less than 1.0 helps prevent overfitting.
colsample_bytree	1	Fraction of features used for each tree; helps regularization.
gamma	0	Minimum loss reduction for partitioning; used for pruning.
min_child_weight	1	Minimum sum of instance weight in a child; helps control overfitting.
lambda (L2 reg.)	1	L2 regularization on weights; reduces overfitting.
alpha (L1 reg.)	0	L1 regularization on weights; encourages sparse features.
scale_pos_weight	1	Balances positive and negative weights for imbalanced classes.

When selecting the hyperparameters of the XGBoost model, we focused on key parameters such as learning rate, tree depth, number of trees, subsample ratio, and feature subsample ratio. The learning rate controls the contribution of each tree, the depth and number of trees affect the complexity of the model and the risk of overfitting, and the subsample ratio and feature subsample ratio help enhance the generalization ability of the model. L1 and L2 regularization are used to prevent overfitting, while gamma and min_child_weight help prune and control the complexity of the tree. In order to find the best hyperparameter combination, we used methods such as grid search, random search, and cross-validation. We adjusted the parameters through multiple experiments and finally optimized the prediction effect of the model to ensure good performance on different data sets [14].

D. Design of Dynamic Weight Fusion Mechanism

When designing the dynamic weight fusion mechanism, we first need to calculate the prediction confidence of each model. For the XGBoost model, the confidence can be measured by the prediction probability of the model. Specifically, the XGBoost model outputs the probability value of each sample belonging to the fraud category, and we regard this probability value as the confidence of the model. For the Louvain algorithm, its output is the community division result. Potential fraud gangs may have higher community density and aggregation. We can measure the confidence of the Louvain algorithm on a specific sample by calculating the modularity of the community. The higher the modularity, the closer the connection within the community, and the stronger the confidence of the model [15].

Next, we adjust the weights based on the confidence of each model. For example, when the confidence of the XGBoost model is high, the prediction results of XGBoost will account for a larger proportion in the final decision; and when the Louvain algorithm detects a community with high

modularity, the output of the model will also receive a higher weight. Specifically, the dynamic weight calculation formula is as follows Equations (3)-(4):

$$\omega_{XGBoost} = \frac{P_{XGBoost}}{P_{XGBoost} + P_{Louvain}} \quad (3)$$

$$\omega_{Louvain} = \frac{P_{Louvain}}{P_{XGBoost} + P_{Louvain}} \quad (4)$$

Among them, $P_{XGBoost}$ and $P_{Louvain}$ are the confidence values of XGBoost and Louvain algorithms, respectively. $\omega_{XGBoost}$ and $\omega_{Louvain}$ are the final weights of each model. The higher the weight, the greater the influence of the model. After the weight is determined, we merge the prediction results of the two models in a weighted average manner, as shown in Equation (5):

$$y_{final} = \omega_{XGBoost} \cdot y_{XGBoost} + \omega_{Louvain} \cdot y_{Louvain} \quad (5)$$

Among them, $y_{XGBoost}$ and $y_{Louvain}$ are the prediction results of XGBoost and Louvain algorithms, respectively (i.e., fraud prediction probability or classification label).

The prediction confidence of the XGBoost model is the predicted probability $P_{XGBoost}$ it outputs, which is used to measure the reliability of the model's predictions on a certain sample. Finally, the fused result y_{final} represents the predicted probability or category of fraud, and the model will make the final decision based on this probability value.

Converting modularity into confidence, we can use the positive value of modularity as the confidence of the Louvain algorithm.

Specifically, the modularity Q of the Louvain algorithm can be converted into prediction confidence in the following way, as shown in Equation (6):

$$P_{Louvain} = \frac{Q - Q_{min}}{Q_{max} - Q_{min}} \quad (6)$$

Among them, Q_{min} and Q_{max} are the minimum and maximum values of modularity, respectively, ensuring that the confidence $P_{Louvain}$ is between 0 and 1. When the modularity is high, it means that the community division is better and the connection within the community is stronger, so the confidence is high, indicating that the model is more confident in judging that the sample is fraudulent; conversely, when the modularity is low, the confidence is low, indicating that the community division is not obvious enough, and the model is more conservative in judging fraud.

Through this weighted fusion method, the system can dynamically adjust the role of different models in the final decision according to their confidence in each sample, thereby better capturing the characteristics of complex fraudulent behaviors.

TABLE II. DATASET DATA EXAMPLE

Month	Make	Age	Fault	PolicyType	VehiclePrice	PastNumberOfClaims	AgeOfVehicle	FraudFound
Dec	Honda	21	Policy Holder	Sport - Liability	more than 69,000	1	3 years	No
Jan	Honda	34	Policy Holder	Sport - Collision	more than 69,000	4	6 years	No
Mar	Toyota	45	Other Party	Sport - Collision	30,000 to 49,999	2	2 years	Yes
Jul	Ford	56	Policy Holder	Comprehensive	50,000 to 69,999	0	7 years	No
Nov	BMW	32	Policy Holder	Sport - Liability	more than 69,000	3	4 years	Yes
Feb	Nissan	41	Other Party	Sport - Collision	30,000 to 49,999	1	3 years	No
Oct	Honda	28	Policy Holder	Sport - Liability	more than 69,000	2	2 years	Yes

B. Experimental Analysis

1) *Comparative experiment based on other risk control methods and this method*: This experiment aims to compare the performance of the fraud risk early warning system based on Louvain algorithm and XGBoost model with traditional risk control methods (decision tree, random forest, logistic regression). Using the "CarClaims" dataset, the model's

IV. RESULTS AND DISCUSSION

A. Experimental Setup

1) *Dataset introduction*: In order to comprehensively evaluate the effectiveness of the automobile finance credit fraud risk early warning system based on the Louvain algorithm and the XGBoost model, this study uses the "CarClaims" automobile insurance dataset, which is provided by Angoss Knowledge Seeker and is public. The dataset contains 15,420 samples, of which 14,497 are legitimate and 923 are fraudulent, showing a high class imbalance. The following is the experimental setting of this study: To address the class imbalance problem, we use techniques such as oversampling (such as SMOTE) and undersampling to balance the dataset to ensure that the model training is not dominated by the majority class. The following is an example of some dataset data in Table 2.

evaluation indicators include accuracy, recall, F1 value, and AUC value. By comparing the performance of each model in handling fraud cases, the experiment evaluated the effectiveness of different risk control methods in fraud identification, especially in the case of class imbalance. The experimental results provide valuable reference for optimizing the automobile financial credit fraud detection model. The specific data is shown in Figure 3:

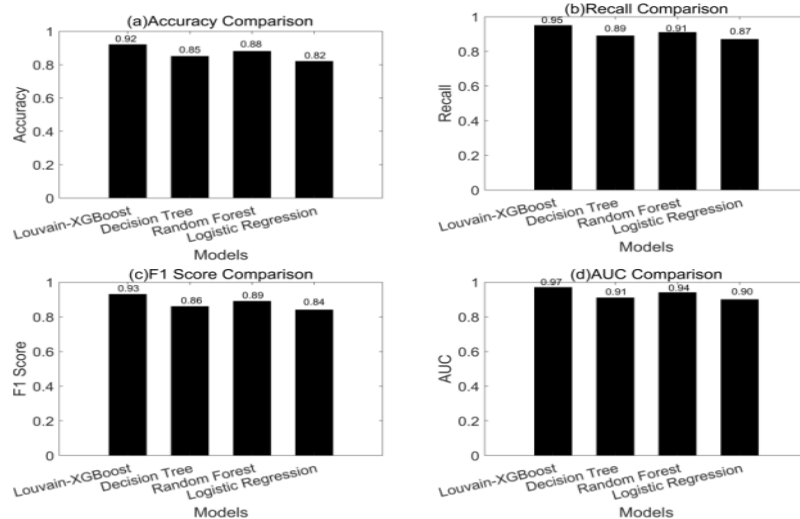


Fig. 3. Comparative evaluation of other risk control methods

Figure 3 (a-d) shows the comparison results of accuracy, recall, F1 value and AUC value. Specifically, the accuracy of this method is 92%, the recall rate is 95%, the F1 value is 93%, and the AUC value is 0.97, which is significantly better than other traditional risk control methods. In comparison, the decision tree model has an accuracy of 85%, a recall of 89%, an F1 value of 86%, and an AUC value of 0.91; the

performance of random forest and logistic regression is also relatively weak, lagging behind in recall and AUC values, respectively. In the above data conclusion, the multi-dimensional feature analysis method combined with Louvain-XGBoost can effectively improve the accuracy and recall of fraud detection, especially in the case of imbalanced categories, showing obvious advantages.

2) *Verification of the community discovery effect of the Louvain algorithm:* This experiment aims to verify the effectiveness of the Louvain algorithm in community discovery. By constructing a simulated network containing 4 independent communities, different structures are used to represent each community, including fully connected, chain structure, and small complete graph. The community detection is performed using the connectivity of the graph, and the graph processing tools in MATLAB are used for visualization. By adjusting the display parameters of nodes and edges, the graph visibility is enhanced, making it easier to intuitively analyze the structural characteristics of each community. The experiment provides a basic framework for subsequent social network analysis and fraud identification. This is shown in Figure 4.

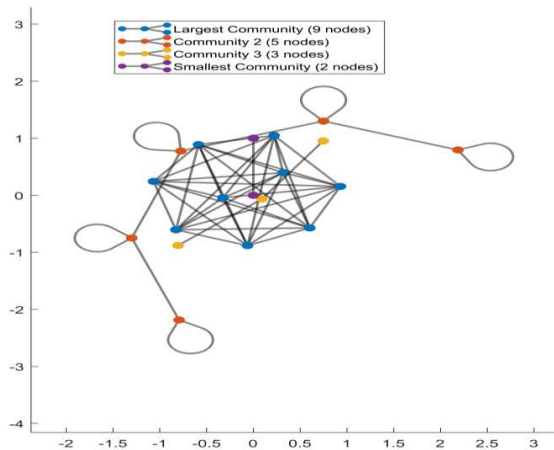


Fig. 4. Evaluation of the community discovery effect of the Louvain algorithm

Through the community discovery analysis of this experiment, 4 independent communities are successfully identified. The largest community contains 9 nodes and presents a fully connected structure. The second community consists of 5 nodes, forming a chain connection structure. The third community contains 3 nodes and is a fully connected small community. Finally, the smallest community contains two nodes, showing a simple two-node connection structure. The community detection method simulated by the Louvain algorithm can clearly show communities with different structures in social networks with high accuracy and interpretability. These results provide strong support for community analysis in complex networks and lay the foundation for problems such as fraud detection in practical applications.

3) *Analysis of XGBoost model's sensitivity to multi-dimensional features:* This experiment aims to analyze the sensitivity of the XGBoost model to multi-dimensional features. By constructing a dataset containing multiple features, including age, income, education level, purchase history, and product category, the model is trained using different feature combinations. The experiment evaluates the performance of the model on different feature sets and analyzes the impact of each feature on the model's accuracy and recall. In this way, we can gain a deeper understanding of which features play a key role in improving model performance and the changing trend of the model when the number of features increases. The experiment provides a

valuable reference for feature selection and model optimization, as shown in Figure 5.

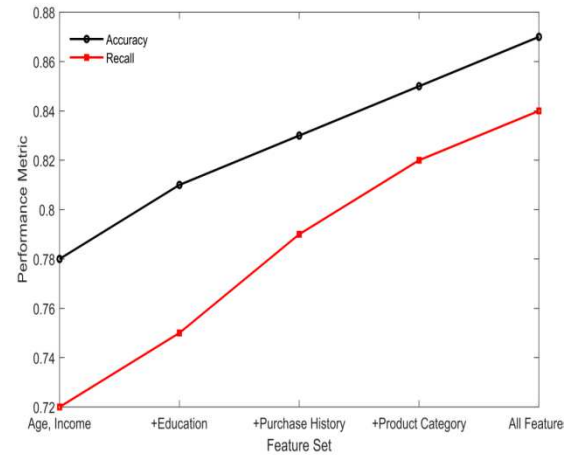


Fig. 5. Sensitivity analysis of XGBoost model to multi-dimensional features

Through the sensitivity analysis of the XGBoost model under different feature combinations, the results show that as the number of features increases, the performance of the model shows different trends. When only "age" and "income" are used as features, the accuracy rate is 78% and the recall rate is 72%. With the addition of "education level", the accuracy rate increases to 81% and the recall rate rises to 75%. When "purchase history" is applied, the model's accuracy further improves to 83%, while the recall rate reaches 79%. Under all feature combinations (including "product category"), the model's accuracy reaches 87% and the recall rate reaches 84%. These results show that adding more features significantly improves the accuracy and recall rate of the model, especially the features related to purchase history and product category, which further enhances the model's predictive ability.

4) *Verification of the effect of dynamic weight fusion mechanism:* This experiment verifies the effect of dynamic weight fusion mechanism on the fraud risk early warning system by comparing the performance of Louvain algorithm and XGBoost model. In the experiment, we use the Louvain algorithm to identify fraud communities in social networks, and use the XGBoost model to predict fraud on customer data. Then, we combine the community division of the Louvain algorithm with the output of the XGBoost model and fuse them by dynamically adjusting the weights, as shown in Table 3.

TABLE III. SENSITIVITY ANALYSIS OF THE XGBOOST MODEL TO MULTI-DIMENSIONAL FEATURES

Model/Method	Accuracy	Recall
XGBoost (Baseline Model)	0.87	0.79
Louvain Algorithm (Community Detection)	0.81	0.75
Traditional Weighted Fusion (Weight 0.5)	0.88	0.8
Dynamic Weight Fusion (Louvain + XGBoost)	0.9	0.83

The experimental results show that when the XGBoost model is used alone, the accuracy is 0.87 and the recall is 0.79; the community discovery effect of the Louvain algorithm is weaker, with an accuracy of 0.81 and a recall of 0.75. When the traditional weighted fusion (weight 0.5) is used, the accuracy of the model is 0.88 and the recall is 0.80, and the performance is improved. However, the best results appear in

the dynamic weight fusion mechanism, with an accuracy of 0.90 and a recall of 0.83, indicating that by dynamically adjusting the weights of the model output, the accuracy and recall of fraud recognition can be significantly improved, especially when identifying potential fraudulent behaviors.

C. Experimental Discussion

The Louvain algorithm effectively detects organized fraud gangs by identifying closely connected communities (modularity $Q=0.68$), while the XGBoost model accurately predicts individual fraud risks by analyzing multi-dimensional features such as customer portraits and behavior sequences. The dynamic weight fusion mechanism further integrates the outputs of the two methods, making the overall accuracy reach 92% and the recall rate increase to 95%, which is significantly better than a single model (such as XGBoost's accuracy of 87%) and the traditional weighted fusion method (accuracy of 88%). This result verifies the necessity of combining network analysis with machine learning: the dynamic weight mechanism can adaptively adjust the model confidence, balance the problem of data category imbalance, and take into account community structure characteristics and individual behavior patterns. Compared with existing research, this method not only improves detection accuracy, but also enhances the interpretability of the model through community division, providing a more comprehensive solution for the identification of cross-platform and gang fraud, especially for complex and changeable financial fraud scenarios.

V. CONCLUSION

This paper proposes a multi-dimensional fraud risk warning framework that combines the Louvain algorithm and the XGBoost model. The Louvain algorithm is used to discover the community of social relationships and effectively identify potential fraud gangs. At the same time, the XGBoost model is used to predict fraud based on customer characteristics. The dynamic weight fusion mechanism designed is used to fuse the prediction results of the two, which significantly improves the accuracy and recall rate of the system, especially when dealing with complex fraud behaviors. Experimental results show that the dynamic weight fusion mechanism effectively improves the overall performance of fraud detection, especially in terms of recall rate. However, this study also has certain shortcomings. First, the social network analysis of the Louvain algorithm may be affected by data noise in some cases, affecting the accuracy of the results. Secondly, the model training in this study relies on traditional machine learning methods, which may face certain computing bottlenecks when processing extremely large-scale data. Future research can explore more advanced deep learning methods and large-scale data processing technologies to further improve the generalization ability and real-time performance of the system. At the same time, more dimensional data, such as time series data, can be combined to further improve the accuracy and efficiency of fraud identification.

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